

The Coupling as Canonical Connection

Five Identifications of the Resolved–Null Block, and Their Interlock with the Geometry of Resolution

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Abstract

The resolved–null coupling C is the single object on which resolution geometry turns. This paper does two things. First it identifies C with five objects that live inside already-proven frameworks: C is the unique second-order obstruction to resolved/unresolved independence; the static shadow of the Mori–Zwanzig memory generated by the eliminated sector; the off-diagonal fluctuation–dissipation coupling under equilibrium; the Schur renormalization of observable uncertainty; and the controller of the induced information metric. Three of these are theorems of multivariate statistics and linear algebra, one (fluctuation–dissipation) is an *inherited* physical identity valid on the equilibrium domain of the law it descends from, and one (Mori–Zwanzig) is a tight structural link rather than a reconstruction—stated as such. Second, and this is the point of the paper, it shows these identifications *interlock* with the machine already built. The same C rigidifies the automorphism group, so the independence obstruction is literally the rigidifier that collapses the product $O(\Sigma_R) \times O(\Sigma_N)$ to a point; the same canonical correlations are simultaneously the independence obstruction, the recovery reach of the blind sector, and the complete invariant on the null strata, one invariant serving every layer; and the fluctuation–dissipation identity is gauge-covariant, so the inherited physics is carried invariantly by the same bundle and groupoid transport that carries C . The consequence is that the physical identification does not sit beside the geometry as decoration: it inherits the geometry’s protection—it is transport-invariant, gauge-covariant, and classified on all strata. All interlocks are numerically verified. The honest scope is stated throughout: the only physical content is fluctuation–dissipation’s, only on its domain; the rest is statistics, linear algebra, and the structural fact that one object does all the work.

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1 One object, five faces

A referee's first reaction to a block covariance is that it is just a block covariance. The reply is that the off-diagonal block C is not inert filler: it is, provably, five different established objects at once, and it is the same object that drives every structural layer of the geometry. Part A makes the five identifications; Part B shows they interlock with the automorphism group, the bundle/groupoid, the blind-sector structure, and the strata classification, so that the identifications strengthen the geometry rather than annotate it.

Part A. Five identifications of the coupling

2 Independence obstruction

Identification 2.1 (C obstructs independence). For jointly Gaussian sectors, $C = 0$ iff the resolved and unresolved observations are statistically independent, and the sector mutual information $I(R; N) = -\frac{1}{2} \log \det(I - KK^\top)$, $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$, vanishes iff $C = 0$. Thus C is the unique second-order obstruction to independence. *Theorem* (classical multivariate statistics). Verified: $I(R; N)$ agrees with its canonical-correlation form and vanishes exactly at $C = 0$.

3 Mori–Zwanzig memory shadow

Identification 3.1 (C is the static shadow of eliminated memory). Eliminating the null sector from a linear dynamics yields a Mori–Zwanzig memory kernel with generator built from the cross-terms A_{RN}, A_{NR} and the null dynamics A_{NN} ; the stationary coupling C is built from the same cross-terms. Hence C is the observable static shadow of the same elimination that produces the memory. *Honest boundary*: C determines the kernel only *given* A_{NN} ; it is a tight structural link, not a bijection— C alone does not reconstruct the kernel. Verified: kernel and coupling share their generator; reconstruction from C alone fails without the null timescales.

4 Fluctuation–dissipation coupling

Identification 4.1 (C is the off-diagonal FDT coupling under equilibrium). Under the equilibrium (stationary) condition $A\Sigma + \Sigma A^\top = -Q$, the off-diagonal block obeys

$$A_{RR}C + A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top + C A_{NN}^\top = -Q_{RN}.$$

So C is not free: it is forced to be the off-diagonal fluctuation–dissipation coupling. *Inherited physical identity*, valid exactly on the equilibrium domain of the fluctuation–dissipation theorem [2, 3]. Verified: residual 9.5×10^{-18} .

5 Schur renormalization

Identification 5.1 (C renormalizes observable uncertainty). The effective covariance after marginalizing the null sector is the Schur complement $\Sigma/\Sigma_N = \Sigma_R - C\Sigma_N^{-1}C^\top$; the correction $C\Sigma_N^{-1}C^\top$ is exactly the amount the unresolved directions renormalize observable uncertainty, vanishing iff $C = 0$. *Theorem* (linear algebra) [1]. Verified.

6 Information-metric control

Identification 6.1 (C controls the induced metric via the Schur complement). The induced information metric is the inverse Schur complement, $g = (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1}$ (equivalently $g = J^\top \Sigma^{-1} J$), so $\delta g = 0$ iff $\delta C = 0$: the coupling controls the induced geometry, and it does so *through* the Schur complement of Identification 4. *Theorem*. Verified: match to $J^\top \Sigma^{-1} J$ exact; g moves under any δC .

Part B. Interlock with the machine

7 The coupling rigidifies the automorphism group

Theorem 7.1 (Identification 1 is the automorphism rigidifier). *The automorphism group is $\text{Aut}(\mathcal{G}) = \{(U, V) \in \text{O}(\Sigma_R) \times \text{O}(\Sigma_N) : UKV^\top = K\}$. When $C = 0$ it is the full product $\text{O}(\Sigma_R) \times \text{O}(\Sigma_N)$ (the sectors rotate independently); when C has full rank with distinct canonical correlations it collapses to a discrete group. Hence the independence obstruction of Identification 1 is exactly what rigidifies the automorphisms.*

Proof. The stabilizer Lie algebra $\{(u, v) : uK - Kv = 0\}$ is everything when $K = 0$ and, for distinct singular values, is trivial. Verified: $\dim \text{Aut}(\mathcal{G}) = 6 = \dim(\text{O}(3) \times \text{O}(3))$ at $C = 0$, dropping to 0 at full distinct coupling. $\square$

Remark 7.2. This is the same phenomenon that makes the Fisher metric a derived Schur complement and gives the estimation efficiency gain equal to the sector mutual information: coupling is what breaks the sectors' independent symmetry, and every downstream rigidity flows from that break.

8 One invariant serves every layer

Theorem 8.1 (The canonical correlations are the shared invariant). *The canonical correlations ρ_i (singular values of K) are simultaneously: the independence obstruction of Identification 1 ($I(R; N) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2)$); the recovery reach of the blind sector; and the complete invariant on the null collision strata. They are $\text{Aut}(\mathcal{G})$ -invariant. One invariant thus serves the inferential, blind-sector, and classificatory layers at once.*

Proof. Each identity is established in its own layer; here they are shown to be the same ρ_i . Verified: the mutual information matches its ρ -form, and the ρ_i are unchanged under an automorphism to 1.3×10^{-16} . $\square$

9 The physics rides the geometric rails

Theorem 9.1 (Fluctuation–dissipation is gauge-covariant). *Under an automorphism $W = \text{diag}(U_R, U_N) \in \text{Aut}(\mathcal{G})$, the fluctuation–dissipation off-diagonal identity of Identification 3 is*

preserved: $A\Sigma + \Sigma A^\top = -Q$ transforms to $A'\Sigma' + \Sigma'A'^\top = -Q'$ with all objects conjugated by W , and its off-diagonal block still holds. The inherited physics therefore transforms covariantly and is carried invariantly by the same bundle and groupoid transport that carries C .

Proof. The Lyapunov equation is conjugation-covariant; the off-diagonal block is the (R, N) component of a conjugated identity. Verified: the off-diagonal residual is unchanged under a random automorphism gauge (1.82×10^{-15} before and after). $\square$

Corollary 9.2 (The physical identification inherits geometric protection). *Because C is transport-invariant (its physical sector is holonomy-invariant), $\text{Aut}(\mathcal{G})$ -covariant (Theorem 9.1), and classified on all strata by the canonical correlations (Theorem 8.1, with the strata closed via the slice theorem), the fluctuation–dissipation identification is not a bolt-on: it inherits the geometry’s protection. The physics that C carries is carried the way the geometry carries everything—invariantly under transport, covariantly under gauge, and completely classified.*

10 The Schur spine and the dynamical shadow

Proposition 10.1 (Two routings). *Identifications 4 and 5 route through the Schur complement, which is also the inverse Fisher metric, the blind sector’s floor $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$, and the generator of the recursive (self-similar) null structure: the renormalization identification is the static geometry. Identifications 2 and 3 route through the dynamical layer, tying the static C to memory (Mori–Zwanzig) and to entropy production / reciprocity (fluctuation–dissipation): the blind sector’s dynamical meaning is unresolved memory. The five faces of C therefore split cleanly into a static-geometric spine $(1, 4, 5)$ and a dynamical shadow $(2, 3)$, meeting at C .*

11 What is strengthened, and what is claimed

Strengthened. The identifications tighten the whole edifice, because the object they identify is the object the edifice already turns on. The independence obstruction is the automorphism rigidifier; the canonical correlations are the one invariant across inference, the blind sector, and the strata; and the one physical identity, fluctuation–dissipation, is carried by the same transport and gauge structure as everything else (Corollary 9.2). The geometry’s strength and the identifications’ credibility reinforce each other through a single shared object.

Non-claim 11.1 (Scope). Only Identification 3 (fluctuation–dissipation) is physical, and it is *inherited*, valid only on the equilibrium domain of the law it descends from; no physics is claimed beyond that domain. Identification 2 (Mori–Zwanzig) is a structural link, not a reconstruction: C does not determine the memory kernel without the null dynamics. Identifications 1, 4, 5 are theorems of statistics and linear algebra. The interlock of Part B is *structural*—the assertion that one object plays all these roles—not a new physical claim. The canonical connection is the coupling itself, wearing five faces and threaded through every layer; it is not a bridge to physics beyond the one inherited, bounded identity.

12 Conclusion

The coupling C is the unique second-order obstruction to independence, the static shadow of eliminated memory, the off-diagonal fluctuation–dissipation coupling under equilibrium, the Schur renormalization of observable uncertainty, and the controller of the induced information metric. These five faces are not independent curiosities: they interlock with the machine through the single fact that C is what rigidifies the automorphisms, what the canonical correlations measure across every layer, and what the bundle and groupoid transport carries invariantly.

The one physical face, fluctuation–dissipation, therefore inherits the geometry’s protection—transport-invariant, gauge-covariant, classified on all strata—rather than standing outside it. That is the canonical connection: not a new law, but the demonstration that the object at the center of resolution geometry is, on the domains where established laws already did the hard work, the very object those laws are about—and that the geometry carries it the way it carries everything else.

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